

The Ukrainian Refugee Influx and Poland's Housing Market: Econometric Evidence, Public Perception, and Policy Lessons

Technical appendix / methodology notes

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A.1 Data and Variables

Data cover the period 2014–2023 (quarterly).

Sources:

- Housing prices (secondary market, average PLN/m², 7 major cities) – Narodowy Bank Polski
- GDP, inflation (CPI) – Główny Urząd Statystyczny
- Interest rates – NBP
- War dummy variable (1 for periods after February 2022, 0 otherwise)

Dependent variable:

- HousePrice – average housing price per square metre (PLN)

Independent variables:

- Inflation (CPI, %)
- GDP (quarterly, adjusted)
- InterestRate (NBP reference rate)
- War (dummy)

A.2 Model and Tests

Model used: ARIMA with external regressors (ARIMAX)

Software: R (packages: forecast, tseries, lmtest, car, orcutt)

Validation checks:

Test	Purpose	Result
Augmented Dickey–Fuller	Stationarity	Data differenced – stationary after transformation
ACF / PACF plots	Autocorrelation	No significant autocorrelation in residuals
Box–Ljung	Autocorrelation (joint)	$p > 0.05 \rightarrow$ no autocorrelation
Shapiro–Wilk	Normality of residuals	$p < 0.05$ (normality violation – noted as limitation)
AIC / BIC	Model fit	Values used for model selection

A.3 Key Figures (from R)

Below are selected outputs from the R analysis.

Figure 1. House Prices in Poland (2014–2024, quarterly) / Boxplot of House Prices

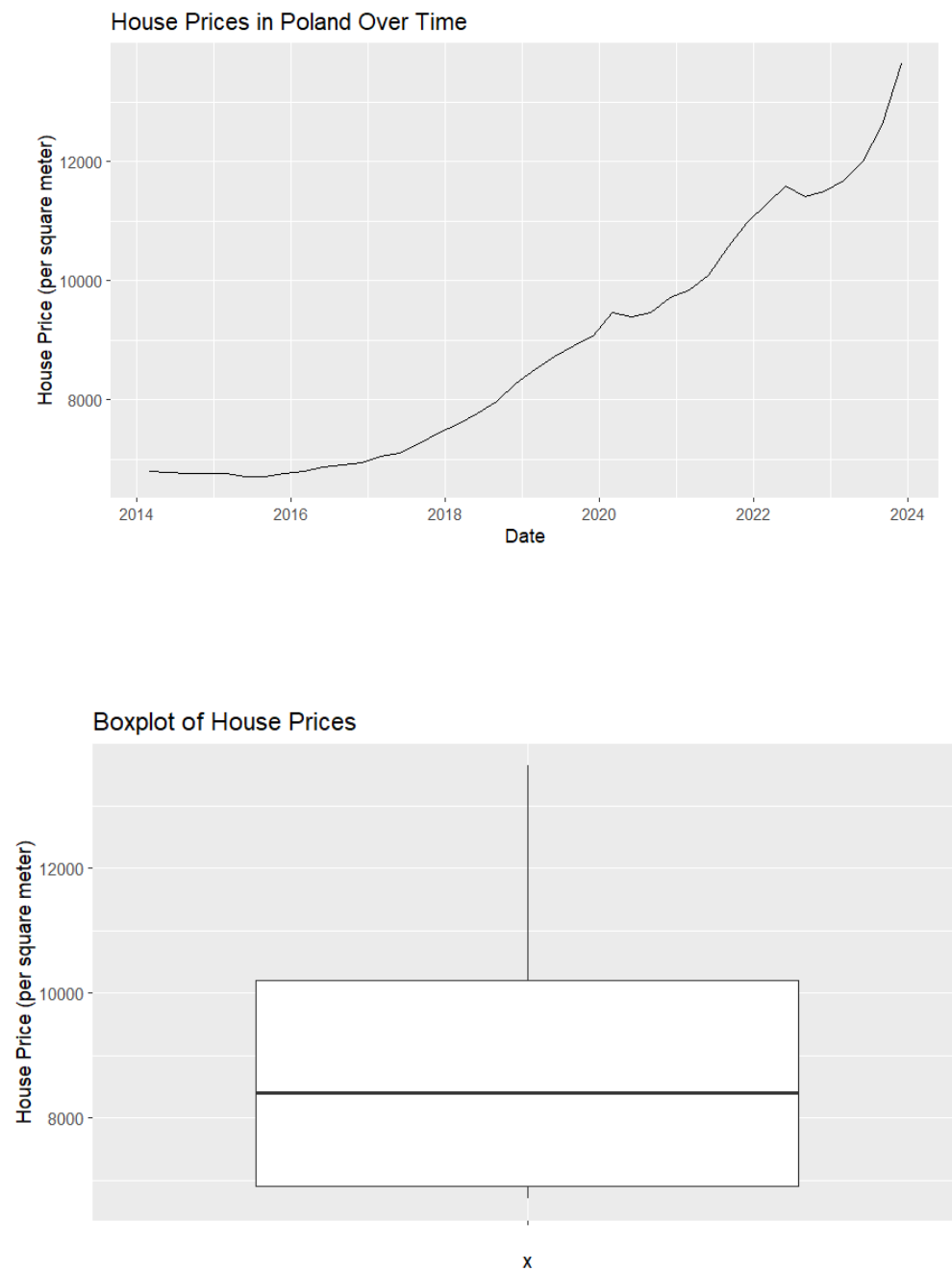


Figure 2. Residuals of ARIMA Model / Histogram of ARIMA Model

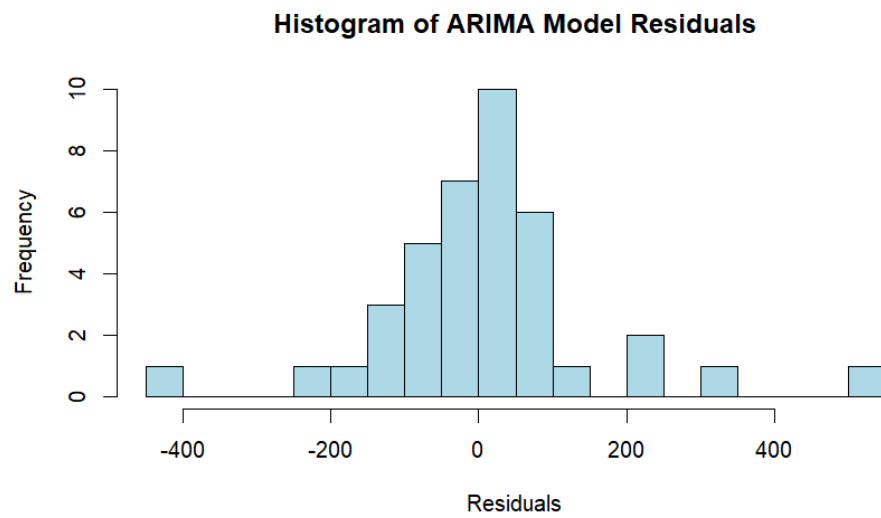
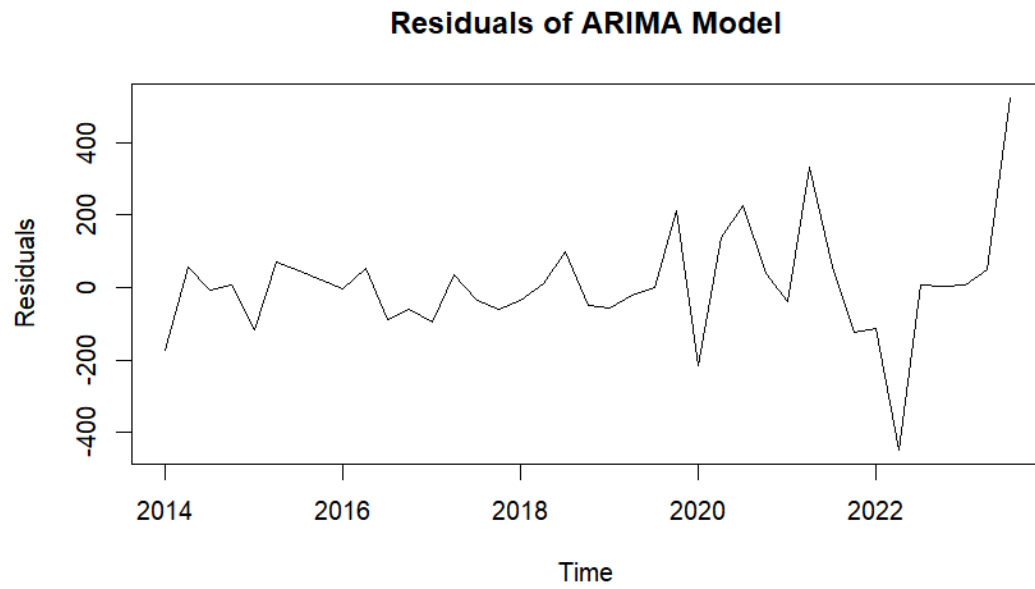


Figure 3. QQ Plot of Residuals

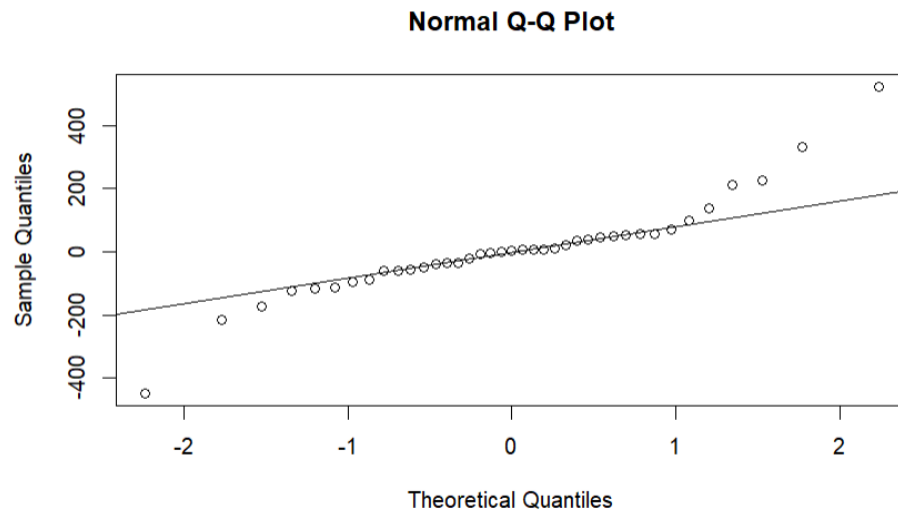
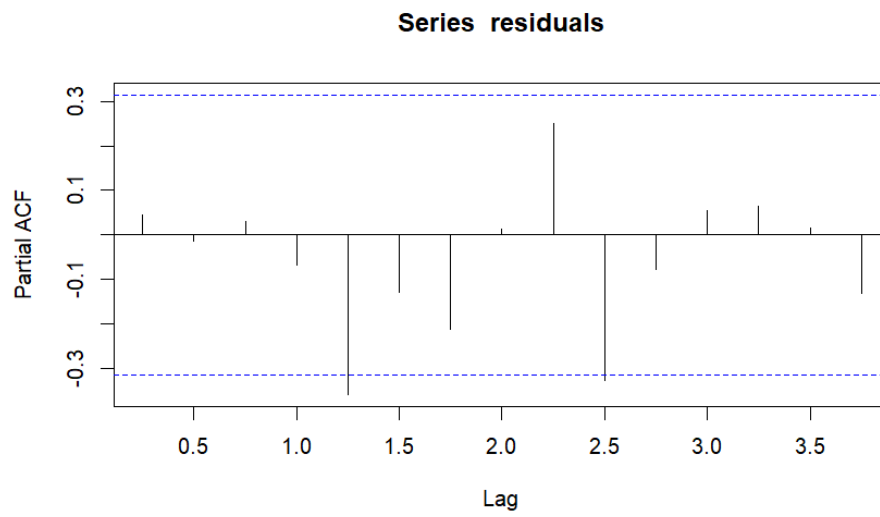
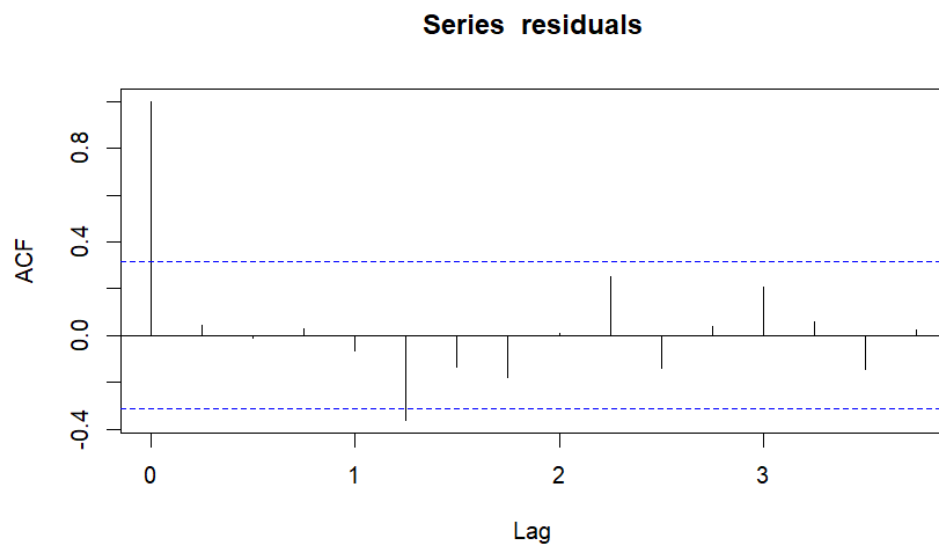


Figure 4. ACF and PACF of Residuals



A.4 ARIMA Model Results

Table 1. ARIMA Model Coefficients (selected output)

Coefficient	Estimate	Std. Error	z value	Pr(> z)	Significant
AR(1)	0.5666	0.1856	3.052	0.00227	Yes
SMA(1)	-0.4244	0.1607	-2.642	0.00825	Yes
Intervention (War)	-16.1	139.05	-0.116	0.9078	No
LogGDP	20.22	19.26	1.049	0.2939	No
LogInflation	42.97	18.46	2.328	0.0199	Yes
LogInterestRate	169.5	50.99	3.324	0.00089	Yes

Model fit:

- AIC = 516.9
- BIC = 528.6
- MAPE (in-sample) = 111.99% (high – indicates model limitations, noted in report)

A.5 Interpretation (brief)

- Inflation and interest rates are statistically significant predictors of housing prices in Poland.
- The war dummy is *not* statistically significant in this model.
- GDP changes did not show a significant effect.
- The model confirms earlier literature that macroeconomic variables (inflation, monetary policy) drive long-term housing price trends.

A.6 R Code and Repository

The full R code used for data cleaning, ARIMA modelling, and diagnostic tests is available at:

 [GitHub repository – insert your link here]

The repository includes:

- Data import and cleaning script
- Stationarity tests (ADF)
- ARIMA model estimation with external regressors
- Residual diagnostics (ACF, PACF, Box–Ljung, Shapiro–Wilk)
- Visualisations (boxplots, residuals, QQ plots)

A.7 Limitations (noted in original report)

- MAPE value is high (111.99%), indicating that the model captures trends but has limited precision in short-term forecasting.
- Normality of residuals was not achieved (noted in Shapiro–Wilk test).
- Some bias may remain due to outliers (COVID-19 period, 2023 data).